Lambda in VPC and Network Access

Explore how to configure AWS Lambda functions for secure access to private VPC resources by associating them with specific subnets and security groups. Understand how Elastic Network Interfaces enable this connection, and learn methods to maintain internet access through NAT gateways while keeping the Lambda function isolated within a private subnet. This lesson helps you build secure, scalable serverless applications that integrate seamlessly with private resources and external services.

A common use case in enterprise applications is a backend service that manages internal employee data. The service can be built using AWS Lambda to process requests and Amazon RDS to store employee records. To keep the database secure, it is usually deployed in a private subnet of a VPC, meaning it is not directly accessible from the public internet. By default, Lambda functions run in an AWS-managed environment that does not have access to this private subnet. As a result, any attempts by the Lambda function to connect to the RDS instance will fail. In this lesson, we'll learn how to provide a Lambda function with access to such AWS resources.

Giving Lambda function access to a custom VPC or private subnet

To allow the Lambda function to communicate with resources that are hosted in private subnets, such as databases, caches, or internal APIs, it must be given access to the same VPC where those resources are located. This setup allows the function to send network requests over private IP addresses, within the boundaries of the AWS environment.

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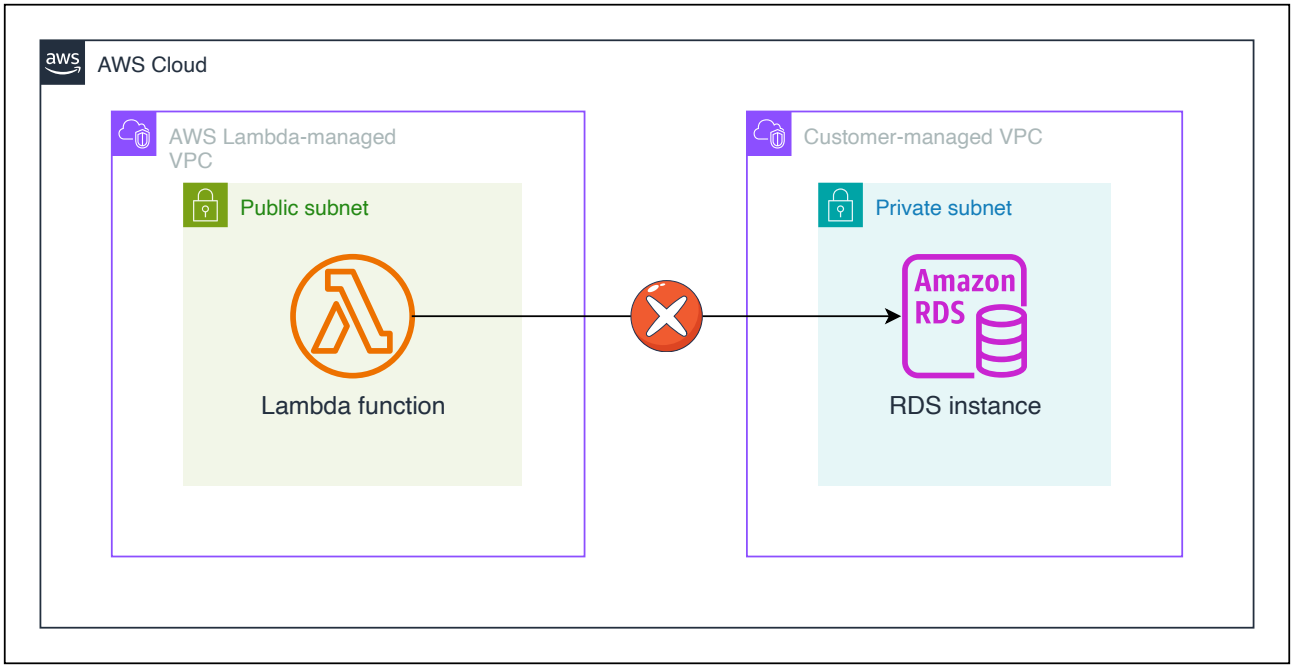

Lambda function connection with a resource within a VPC

To establish this connection, we need to configure the function to use networking components from the target VPC. Specifically, this involves selecting the following within the Lambda function configurations:

- Subnets where the Lambda function should be placed. These are usually private subnets that contain the resources the function needs to access.
- Security groups that control the function's network access. These define which inbound and outbound traffic is allowed.

We can create this association using the AWS Management Console, the AWS CLI, or one of the AWS SDKs. For example, the following commands update a Lambda function's configuration to connect it to a VPC:

AW...

```
aws lambda update-function-configuration \
  --function-name process-employee-data \
  --vpc-config SubnetIds=<Subnet-ID> \
               SecurityGroupIds=<SecurityGroup-ID>
```

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Associate Lambda with a VPC using AWS CLI

This command tells AWS to link the function named `process-employee-data` to the specified subnets and security group. Once connected, the function will be able to reach private resources within those subnets.

Lambda function connection with a resource within a VPC

This association does not move the Lambda function into the VPC. Instead, AWS creates network interfaces inside the selected subnets and links them to the function behind the scenes. These interfaces serve as a secure connection point between the Lambda function and our VPC. This association defines how the Lambda function connects to other AWS resources.

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This setup allows the function to operate securely inside our network boundaries, without requiring public exposure of any sensitive service.

How VPC connectivity affects Lambda's networking behavior

When a Lambda function is connected to a VPC, it no longer uses the default networking environment managed by AWS. Instead, AWS creates Elastic Network Interfaces (ENIs) inside the subnets that were selected during configuration. These ENIs act as private entry points into our VPC and allow the function to communicate with internal services over private IP addresses.

Because the function now uses these ENIs for all its network traffic, it loses the default internet access it would otherwise have. This change happens automatically and is an important part of the function's new behavior. The ENIs are bound to private subnets, and private subnets do not have a route to the internet by default.

Lambda function connection with a resource within a VPC and internet

This change is not always a limitation. If the Lambda function only needs to access other private resources, such as an Amazon RDS database or an internal API, it can operate entirely within the VPC without any issues. However, if the function also needs to access public services, such as third-party APIs or services without VPC endpoints, this loss of internet access will prevent it from functioning as intended.

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To restore internet access while keeping the function in a private subnet, we should add the following components within the VPC:

- A NAT gateway in a public subnet of the VPC to forward internet-bound traffic from private subnets.
- An internet gateway is connected with the public subnet hosting the NAT gateway, which is the actual connection point between the VPC and the internet.
- The private subnet's route table must include a route that sends outbound traffic (typically `0.0.0.0/0`) to the NAT gateway.

> **Note:** To access resources in a VPC, the Lambda execution role must allow `ec2:CreateNetworkInterface`, `ec2:DescribeNetworkInterfaces`, and `ec2:DeleteNetworkInterface`. These permissions let Lambda manage the ENIs required for private subnet connectivity. Without them, VPC access will fail at runtime.

The code below shows how we can set up this architecture to provide internet access to our Lambda function:

AW...

```
# Set the environment variables
VPC_ID="VPC-ID"
PUBLIC_SUBNET_ID="PublicSubnet-ID"
PRIVATE_SUBNET_ID="PrivateSubnet-ID"

# Create Internet Gateway (if doesn't exist)
IGW_ID=$(aws ec2 create-internet-gateway --query 'InternetGateway.Int

# Attach Internet Gateway to the VPC
aws ec2 attach-internet-gateway \
  --internet-gateway-id "$IGW_ID" \
  --vpc-id "$VPC_ID"
```

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Connect Lambda with internet using AWS CLI

Together, this setup allows the Lambda function to use the ENI to reach the NAT gateway. The NAT gateway then sends the traffic through the internet gateway, enabling access to public services—without exposing any private subnets to incoming internet traffic.

Lambda function connection with a resource within a VPC and internet

At the same time, the use of ENIs allows Lambda to behave like any other compute resource in a VPC. Its network access can be tightly controlled using security groups, which define which types of traffic are allowed in and out. For example, a database hosted in the same VPC may allow connections only from the Lambda function's security group, creating a secure and isolated communication channel.

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So by carefully associating Lambda functions with the correct subnets and security groups, developers can ensure that their serverless applications operate

securely within a VPC while maintaining the ability to access necessary resources. Proper network design and security group management are essential to building reliable, scalable, and secure Lambda-based architectures.

Conclusion

By associating a Lambda function with a VPC, we can allow it to access resources in private subnets, such as RDS databases. If the function also needs internet access, it must route traffic through a NAT gateway in a public subnet. That public subnet, in turn, must be connected to an internet gateway to enable outbound connectivity. This setup allows secure access to internal AWS resources while still supporting internet-bound requests.

Mark As Completed

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